

Laser beams from Earth have been **reflected** back from a panel left on the **Moon** by astronauts.

LIGHT POWER

LASERS

In 1960, American scientist Theodore Maiman built a device that could emit a narrow, powerful beam of single-color light—the world's first laser. The name stands for “light amplification by stimulated emission of radiation.” In a laser device, a material is supplied with energy. This makes the atoms of the material release energy in the form of light waves that travel as a straight beam of light. The properties of laser light give it many applications, such as in surgery, telescopes, and communications.

LASER LIGHT

Light is a form of energy that travels as waves (see page 184). Light waves from most sources have a mix of different wavelengths and colors that move out of step. However, waves of laser light have identical wavelengths and travel perfectly in step with each other. These properties mean laser light is very intense and its energy can be easily concentrated onto specific objects.

